Experiment 05

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Aim

To perform task analysis for an e-commerce application and design a user flow diagram that represents the steps a user follows while purchasing a product online.

Task Analysis

Task Analysis is the process of breaking down user interactions into smaller steps to understand how users complete a specific goal.

In this experiment, the goal of the user is to successfully **purchase a product through an e-commerce application**.

This task analysis helps to:

• Understand user actions during online shopping.
• Identify decision points such as login and payment method.
• Improve the efficiency of the purchasing process.
• Reduce user confusion during checkout.

Key Aspects of Task Analysis

• Identifying user goal – Purchasing a product online.
• Understanding the sequence of actions from opening the app to order confirmation.
• Recognizing decision points (Login/Register, Payment Method).
• Ensuring smooth navigation between steps.

User Flow

User Flow is a visual representation of the path users take to complete a task within an application. In this design, the user flow represents the **complete process of purchasing a product in an e-commerce application from start to end**.

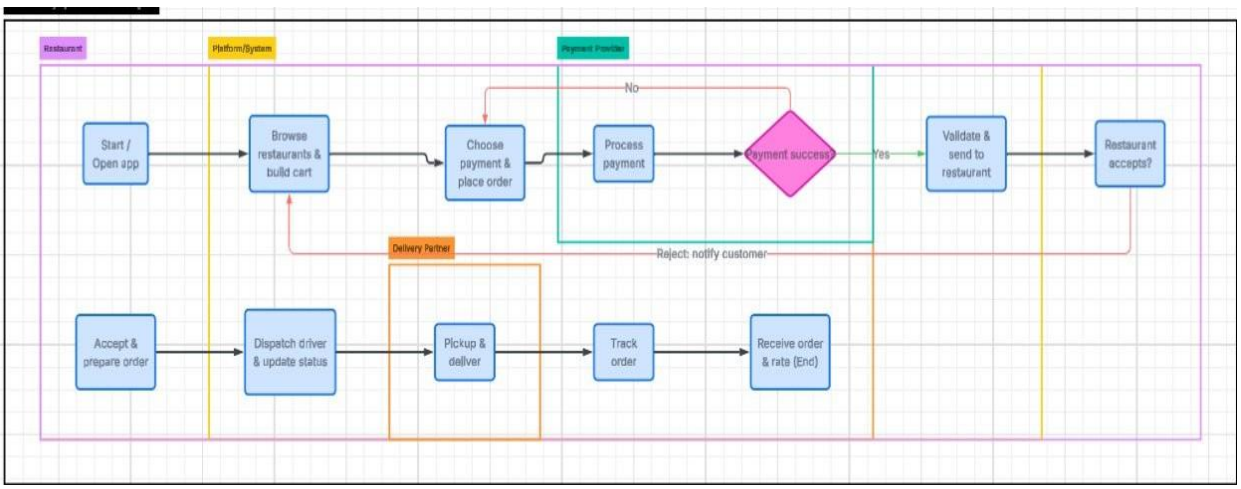

Figure 1: E-Commerce User Flow Diagram

User Flow for E-Commerce Shopping System

Start → Open App → Login/Register → Enter Username & Password → Login Successful → Home Screen → Browse Products → Select Category → View Product List → View Product Details → Add to Cart → Open Cart → Checkout Process → Enter Shipping Details → Select Payment Method → Payment Verification → Payment Success → Order Confirmation → Track Order → End

Step-by-Step Explanation

Step 1: Start
The user opens the e-commerce application.

Step 2: Login or Register
If the user already has an account, they log in. If the user is new, they must register before continuing.

Step 3: Home Screen
After successful login, the user is redirected to the home screen. The user can browse products or search for specific items.

Step 4: Browse or Search Products
The user selects a product category or searches for a product. The system displays a list of available products.

Step 5: View Product Details
The user clicks on a product to view details such as product description, price, images, and specifications.

Step 6: Add to Cart
If the user wants to purchase the product, they add it to the shopping cart. The cart stores selected products for checkout.

Step 7: Open Cart and Review
The user opens the cart to review selected items. They can check quantity and product details.

Step 8: Proceed to Checkout
If everything is correct, the user proceeds to checkout.

Step 9: Enter Shipping Details
The user enters delivery information such as address, phone number, and delivery preferences.

Step 10: Payment Method Decision
The user selects a payment method: Online Payment or Cash on Delivery (COD).

Step 11: Payment Verification
If online payment is selected, the system verifies the payment transaction. After verification, the payment is confirmed.

Step 12: Order Confirmation
The order is successfully placed. The system displays an order confirmation message.

Step 13: Track Order
The user can track the order status through the application.

Step 14: End
The shopping process is completed successfully.

Decision Points in Flowchart

• Login / Register decision
• Add to Cart decision
• Payment Method decision
• Payment Verification decision
These decision nodes allow flexible navigation and ensure smooth user interaction within the e-commerce system.

Wireframes

Wireframes are low-fidelity visual representations of the application's user interface. They focus on layout, structure, and user interactions without detailed design elements.

1. Login Screen Wireframe

The login screen allows users to authenticate and access the application. It includes email/username input, password input, login button, and options for password recovery and account creation.

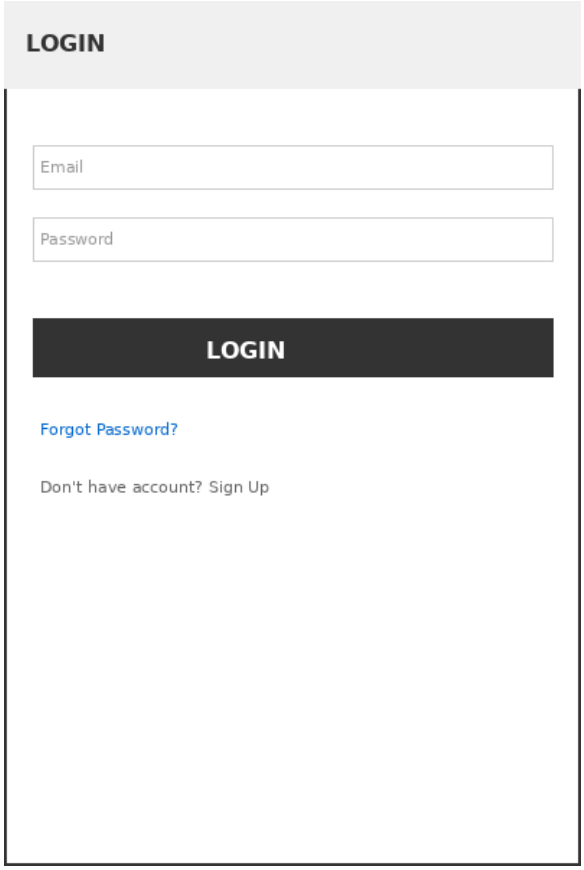

Figure 2: Login Screen Wireframe

2. Home Screen Wireframe

The home screen serves as the main dashboard where users can explore products. It includes a search bar, featured products section, latest insights, and a bottom navigation bar for easy access to different sections.

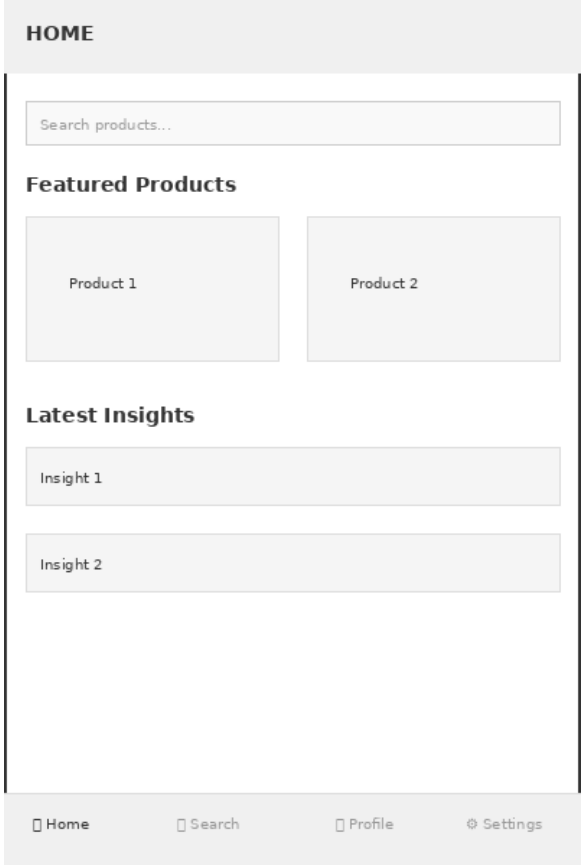

Figure 3: Home Screen Wireframe

3. Checkout Screen Wireframe

The checkout screen allows users to enter shipping details and select a payment method. It includes form fields for address information, payment method selection, and a continue button to proceed with the order.

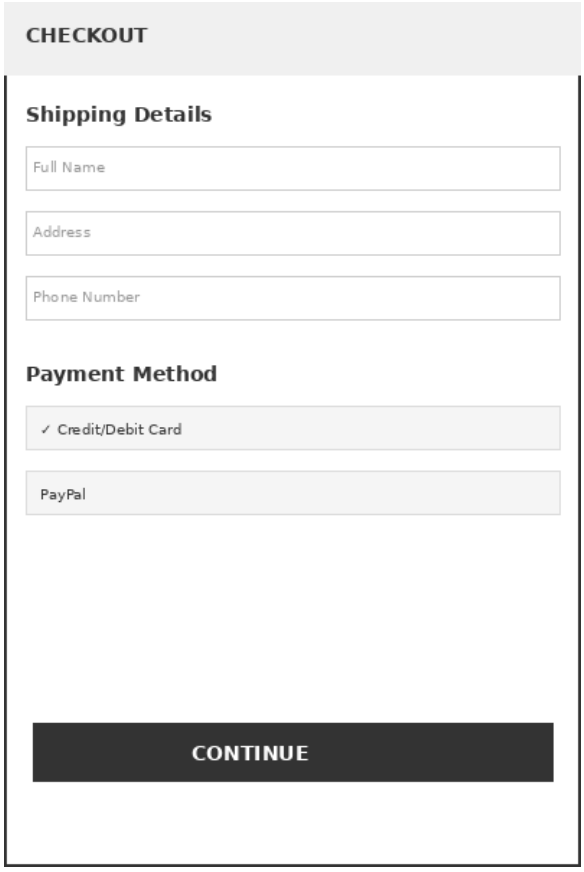

Figure 4: Checkout Screen Wireframe

Result

The task analysis and user flow diagram were successfully created for the e-commerce application. The diagram clearly shows the sequence of steps and decision points involved in the online shopping process, from opening the application to tracking the order. The wireframes provide a clear visual representation of the key screens in the shopping process, demonstrating the layout and structure of the user interface.